

MATH Unit 2 Question Bank — Differential Calculus

Mathematics (MATH) · Unit 2 · Differential Calculus · 2025

Important and frequently asked questions, grouped by marks.

Part A — Short answer (2 marks)

1. Compare Mean Value Theorems with Lagrange Multipliers. When would you prefer one over the other?
2. Compare Taylor and Maclaurin Series with Mean Value Theorems. When would you prefer one over the other?
3. Explain Partial Derivatives with a neat diagram and a suitable example.
4. Write short notes on Jacobians.
5. Write short notes on Maxima and Minima.
6. Compare Lagrange Multipliers with Taylor and Maclaurin Series. When would you prefer one over the other?

Part B — Long answer (10 marks)

1. Illustrate Mean Value Theorems with a worked example.
2. Compare Taylor and Maclaurin Series with Maxima and Minima. When would you prefer one over the other?
3. Discuss the role of Partial Derivatives in Differential Calculus.
4. Illustrate Jacobians with a worked example.
5. What is meant by Maxima and Minima? Discuss its advantages and limitations.
6. What is meant by Lagrange Multipliers? Discuss its advantages and limitations.
7. Write short notes on Mean Value Theorems.
8. Illustrate Taylor and Maclaurin Series with a worked example.
9. Write short notes on Partial Derivatives.
10. Write short notes on Jacobians.
11. Illustrate Maxima and Minima with a worked example.
12. Discuss the role of Lagrange Multipliers in Differential Calculus.